



Matrix Theory

14 Projections onto lines

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- Zero dot product $\rightarrow$ orthogonal

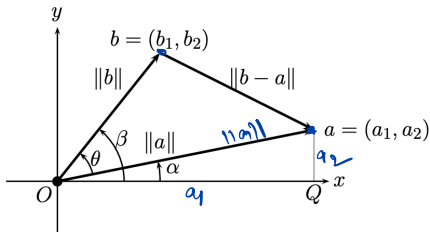


- Zero dot product $\rightarrow$ orthogonal
- What about arbitrary angles?



- Zero dot product $\rightarrow$ orthogonal
- What about arbitrary angles?
- Also, how to find the distance from a point to a given subspace?

Inner product and Angle



$$\sin(\alpha) = \frac{a_2}{\|a\|} \quad \checkmark$$

$$\cos(\alpha) = \frac{a_1}{\|a\|} \quad \checkmark$$

$$\cos(\theta) = \cos(\beta - \alpha) \quad \checkmark$$

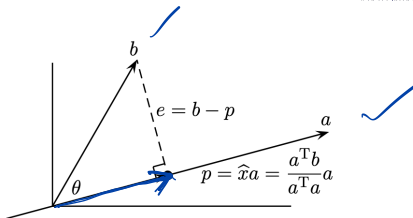
$$= \cos(\beta)\cos(\alpha) + \sin(\beta)\sin(\alpha) \quad \checkmark$$

$$\cos \theta = \frac{a_1 b_1 + a_2 b_2}{\|a\| \|b\|} = \frac{a^T b}{\|a\| \|b\|} \quad \checkmark$$

$$\underline{\underline{= 1}}$$

$\theta = 0^\circ$ (↗)

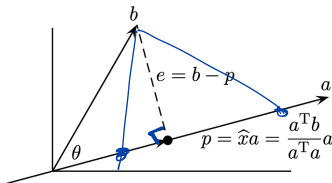
Projection onto a line



Projection p of b onto a

- Must be some multiple of a , i.e., $p = \underline{\hat{x}a}$

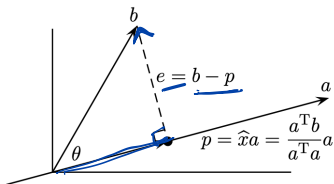
Projection onto a line



Projection p of b onto a

- Must be some multiple of a , i.e., $p = \hat{x}a$ ✓
- p is the closest point to b on a

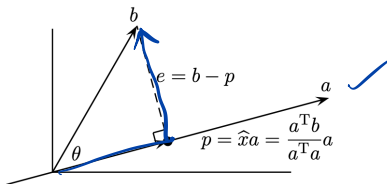
Projection onto a line



Projection p of b onto a

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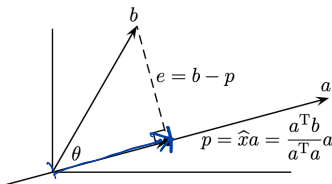
Projection onto a line



Projection p of b onto a

- Must be some multiple of a , i.e., $p = \hat{x}a$
- p is the closest point to b on a
- e must be orthogonal to a
- $(b - \hat{x}a) \perp a$, i.e., $\underline{a^T(b - \hat{x}a) = 0}$

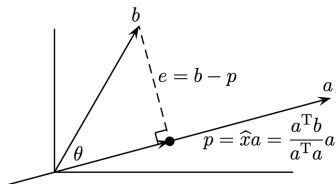
Projection onto a line



Projection p of b onto a

- Must be some multiple of a , i.e., $p = \hat{x}a$
- p is the closest point to b on a
- e must be orthogonal to a
- $(b - \hat{x}a) \perp a$, i.e., $a^T(b - \hat{x}a) = 0$ ✓
- $\hat{x} = \frac{a^T b}{a^T a}$ and $p = \hat{x}a = \frac{a^T b}{a^T a}a$

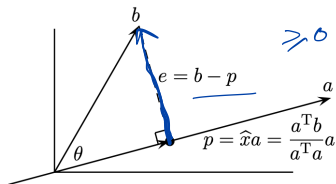
Schwarz inequality



Projection p of b onto a

- Length of e is non-negative

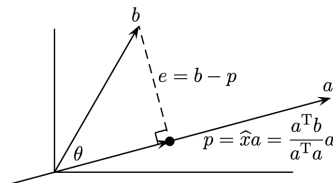
Schwarz inequality



Projection p of b onto a

- Length of e is non-negative
- $\|e\|^2 = \|b - p\|^2 \geq 0$

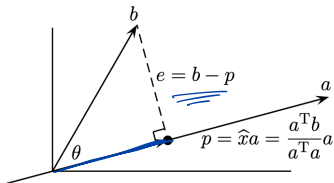
Schwarz inequality



Projection p of b onto a

- Length of e is non-negative
- $\|e\|^2 = \|b - p\|^2 \geq 0$
- $\|b - \frac{a^T b}{a^T a} a\|^2 = b^T b - 2 \frac{(a^T b)^2}{a^T a} + \left(\frac{a^T b}{a^T a}\right)^2 a^T a = \frac{(b^T b)(a^T a) - (a^T b)^2}{a^T a} \geq 0$

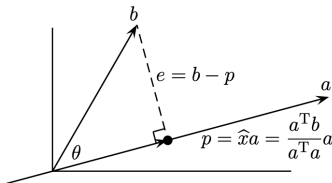
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- $(b^T b)(a^T a) \geq (a^T b)^2$

Schwarz inequality



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- $(b^T b)(a^T a) \geq (a^T b)^2$

Schwarz inequality: $|a^T b| \leq \|a\| \|b\|$



Schwarz inequality



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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$$|a^T b| \leq \|a\| \|b\|$$

- Equality holds iff b is a multiple of a

Schwarz inequality



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$$|a^T b| \leq \|a\| \|b\|$$

- Equality holds iff b is a multiple of a
- Also known as the Cauchy-Schwarz-Buniakowsky inequality

Can projection be represented as a matrix/transformation?



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- $p = \hat{x}a$

Can projection be represented as a matrix/transformation?



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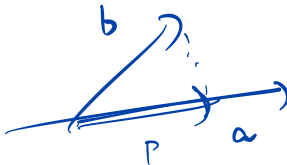
- Let's rewrite: $p = \underbrace{a \frac{a^T b}{a^T a}}_{1 \times 1}$

$$\begin{array}{c} \underbrace{n \times 1} \quad \underbrace{1 \times n} \\ \underbrace{n \times n} \quad n \times 1 \\ n \times 1 \end{array}$$

$$\begin{array}{c} \boxed{\begin{array}{c} a \cdot a^T \\ \hline a^T a \end{array}} b \end{array}$$

$$\frac{a^T b \cdot a}{a^T a}$$

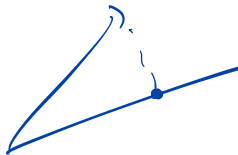
$$p = \frac{P \cdot b}{1}$$



Can projection be represented as a matrix/transformation?



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- $p = \hat{x}a$
- Let's rewrite: $p = a \frac{a^T b}{a^T a}$
- $p = Pb$, where $P = \frac{aa^T}{a^T a}$

$$P = \frac{aa^T}{a^T a} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

Can projection be represented as a matrix/transformation?



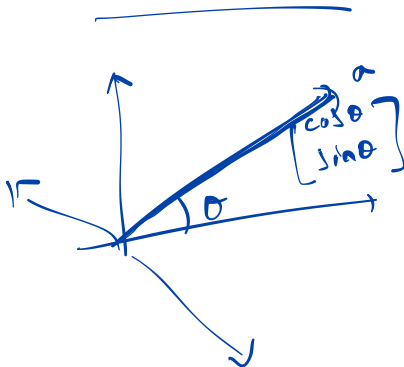
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- $p = \hat{x}a$
- Let's rewrite: $p = a \frac{a^T b}{a^T a}$
- $p = Pb$, where $P = \frac{aa^T}{a^T a}$
- P is a rank-1 matrix, symmetric, and $P^2 = P$

Example



- Project onto a line at θ angle to x-axis



$$P = \underline{a} \underline{a}^T$$

$$\begin{bmatrix} c & s \end{bmatrix}$$

$$= \begin{bmatrix} c^2 & sc \\ cs & s^2 \end{bmatrix}$$



- Projections and Least Squares

Rough Work



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